



Validation of a bioreactor for the growth of black soldier fly larvae: Test with animal feces, *agave* residues and vinasse

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ABSTRACT

For this study, a controlled and monitored bioreactor was constructed to examine the environmental variables, CO₂ behavior during the biodegradation process, aeration requirements, and their effect on the growth of black soldier fly larvae. The larvae were fed with bovine and ovine manure residues, as well as mixtures of *agave* bagasse and vinasse, which are by-products of mezcal distillation, at feeding rates of 47.67, and 95.23 mg/larvae. day. The larvae with the highest growth were those fed on sheep manure, while the larvae with the lowest growth were fed on the mixture of *agave* bagasse and vinasse. The nutritional content of the frass was analyzed, and C:N ratios of 18.75 and 10 were obtained for the fertilizers derived from bovine and ovine manure, respectively, though phosphorus concentrations were low. Data from the sensors validated the monitoring system and data acquisition, enabling the establishment of aeration requirements based on CO₂ concentration and projecting mathematical models to complement this study.

1. Introduction

The sustainable management of agro-industrial and organic wastes has become a global challenge due to the quantity, diversity and complexity of the materials generated. Among these, animal manure, sludge from wastewater treatment plants, distillery residues, brewery by-products, fruit pulps, lignocellulosic bagasse such as *agave* and sugarcane bagasse, and other agricultural residues rich in cellulose, hemicellulose and lignin represent a major environmental impact. Despite efforts to implement traditional techniques such as composting, vermicomposting and direct soil application, these methods face limitations due to the presence of heavy metals, recalcitrant compounds and high operating costs. In this context, the use of insects such as *Hermetia illucens* emerges as an innovative biotechnological alternative, capable of efficiently reducing waste volumes while producing biomass rich in proteins, lipids and other valuable compounds. The incorporation of bioreactors designed to monitor and optimize parameters such as temperature, humidity and gaseous emissions reinforces this strategy, providing a more controlled, efficient and sustainable approach for the valorization of organic waste and the mitigation of its environmental

impacts.

1.1. Production and composition of agroindustrial residues

Manure produced in livestock activities is one of the primary agricultural raw materials for enhancing soil nutrients. The composition of manure depends on the species it comes from and the type of feeding, which varies depending on whether the animals are raised for fattening or production purposes. The feed provided consequently affects the nitrogen, phosphorus, and potassium content of the manure (Khoshnevisan et al., 2021). Environmental conditions also influence the quality of manure due to volatilization losses, which reduce nitrogen concentrations. Manure is often stored in the open air, leading to nutrient losses through leaching (Lee et al., 2021).

A cattle weighing 550 kg can produce up to 90 kg of nitrogen, 18 kg of phosphorus, and 83 kg of potassium per year (Iglesias Martínez, 1995). In China alone, in 2019, 91.4 million head of cattle were reported to have produced 3.8×10^8 tons of manure (Chen et al., 2023). Nitrogen in manure occurs in three forms: mineral, mineralizable organic nitrogen in the first year, and mineralizable organic nitrogen in subsequent

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years. Potentially mineralizable nitrogen refers to soil organic nitrogen that can be converted by aerobic heterotrophic microbial activity into soluble inorganic forms, such as ammonium and nitrate (Galvis-Spinola and Hernández-Mendoza, 2004). Nitrogen concentrations vary by manure type; in poultry manure, ammonium and nitrate content accounts for 70 %, while in cattle and pig manure, it accounts for 40 % and 50 %, respectively (Iglesias Martínez, 1995). During composting processes, between 16 % and 76 % of the nitrogen content is lost (Chen et al., 2023).

1.2. Waste from the distilling industry

Other wastes with valorization potential include those from the alcoholic fermentation process. The distillery industry is among the 17 most polluting industries in the world (Dhote et al., 2021). These industries not only produce effluents with high organic matter concentrations, such as BOD levels between 45 000 and 60 000 mg/L (Mohana et al., 2009), but also generate pressed sludge with no direct industrial application (Fito et al., 2017). In the EU alone, approximately 3.4 million tons of solid waste are produced annually, including yeast and cereal residues, which are used as animal feed ingredients (Aliyu and Bala, 2011). Of the sludge generated, 80 % is disposed of in landfills after dewatering, which poses significant environmental problems. Landfills contribute to soil, air, and water pollution, including the release of harmful gases from untarred sludge (Ferronato and Torretta, 2019).

Distilleries generate three types of solid waste: bagasse, distillate residues that vary depending on the raw materials used (yeast sludge), and wastewater treatment sludge, which is considered non-usable material. In terms of liquid waste, stillages are produced at a rate of 1 to 6 liters for every liter of finished product (S. P. Wang et al., 2022). Additionally, in beer production, wastewater is generated from washing spent malt grains, which also has valorization potential. However, the industry faces resistance in adopting valorization methods. Currently, alternatives include composting, although this has limitations due to the waste's pH and its content of organic substances and metals (Mahaly et al., 2018). Some small companies use these wastes for crop irrigation, leading to soil contamination, while decomposition in landfills contributes to an increased carbon footprint.

1.3. Agave bagasse in mezcal production

In Mexico, a prominent distillery industry is the production of mezcal, made from various species of the genus *Agave*, commonly known as *magüey*. The production of mezcal from species such as *A. angustifolia*, *A. potatorum*, *A. karwinskii*, and *A. marmorata* generates an average of 5.63 to 12.83 kg of bagasse per liter of mezcal, with an average of 9 kg/L depending on the milling and distillation processes (Flores-Ríos et al., 2020). In 2021, 8.1 million liters of mezcal were produced. In Puebla alone, 278 625.93 liters of mezcal were produced in 2022, representing 3.44 % of national production, while bagasse generation reached approximately 4179 tons (Consejo Mexicano and Regulador de la Calidad del Mezcal A.C., 2023). Agave bagasse is a lignocellulosic residue composed of 43 % cellulose, 19 % hemicellulose, and 15 % lignin by dry weight (Saucedo-Luna et al., 2011).

The application of untreated sludge to soil reduces manganese availability, increases soil alkalinity, and inhibits seed germination (Parvathi and Hemanth Kumar, 2014). The Maillard reaction occurs between amino groups and carbonyl groups, leading to the formation of melanoidins a dark brown pigment found in distillery effluent sludge and stillage (Chandra et al., 2008). Melanoidins inhibit the development of beneficial microorganisms in both soil and water.

1.4. Waste treatment methods

Sludge and residues can be treated using physical, chemical,

biological, and thermal methods, including pyrolysis, dehydration, briquetting, and composting, among others. However, the costs associated with these treatments can reduce producers interest. The most promising option is to use these materials for agricultural purposes; however, it is essential to analyze their toxicity, especially concerning their application to food crops (Mahaly et al., 2018).

Composting and vermicomposting are viable treatment options; however, they have limitations due to the presence of heavy metals in distillery sludge. These heavy metals can negatively impact the population of microorganisms and microbial enzymes in vermicomposting (Antizar-Ladislao et al., 2010). Additionally, applying higher concentrations of distillery sludge to the soil can lead to a subsequent decrease in soil fertility.

Another alternative for waste treatment and nutrient valorization is the use of insects. Processed animal protein (PAP) obtained from seven insect species can be utilized as feed for non-ruminant animals, including fish, pigs, and poultry, as approved by the European Commission (Beyers et al., 2023).

1.5. Use of black soldier flies in waste valorization

Hermetia illucens is a dipteran fly belonging to the family *Stratiomyidae*, found in tropical, subtropical, and warm regions of the Americas (Z. Liu et al., 2020). Adult flies do not have stingers, mouthparts, or digestive organs. Males have a golden abdomen, while females are brown in color, with a body size ranging from 15 to 20 mm in length (Kumar et al., 2018). Females mate two days after emerging and can produce between 206 and 639 eggs in a single clutch (Tomberlin et al., 2002).

The maturation process of *Hermetia illucens* takes two months with an efficient substrate source and up to four months in the presence of nutritional deficiencies (Makkar et al., 2014). Under laboratory conditions with temperatures between 27 and 28 °C, development from egg to prepupa takes 22 to 24 days, while the complete development from egg to adult requires 40 to 43 days (Tomberlin et al., 2009). Larvae undergo six instars, corresponding to six molts, before pupation. In the sixth instar, they stop feeding and search for a dry place to transition to the adult stage. The instars can be identified by differences in the width of the head capsule (Giannetto et al., 2017).

Multiple investigations have reported that black soldier fly (BSF) larvae can grow on a wide variety of substrates, including plant materials, animal feces, tannin-containing substrates, and brewery waste, among others (Bonelli et al., 2019). According to Müller and collaborators, BSF larvae produce enzymes that degrade cellulose, chitin, and lignin (Müller et al., 2017), demonstrating their effectiveness in the decomposition of fibrous materials.

One of the concerns regarding the use of insects is the potential proliferation of pathogens. However, the black soldier flies only feeds during its larval stage due to its small oral structure (Li et al., 2017). These larvae require substrates with high moisture content, so they naturally proliferate in environments where manure and other organic remains are present. In this way, they biotransform waste into proteins and lipids, which they subsequently use to develop into adults (Van Huis et al., 2013).

Larvae can weigh up to 220 mg and feed rapidly on diets ranging from 25 to 500 mg of fresh matter per larvae per day (Makkar et al., 2014). Giannetto et al. reported that larvae reached the second larval stage in 4 days, with a mass of 91 mg and a size of 4 mm, while the fifth larval stage was reached in 13 days, with a mass of 220 mg and a length of 14 mm (Giannetto et al., 2017). Studies on the average nutrient composition of *Hermetia illucens* larvae indicate that it is highly dependent on their diet, with approximate compositions of carbohydrates (37 %), chitin (9 %), and lipids (15 %), along with protein content (32 %) in dry mass (Singh and Kumari, 2019). Larval survival to the prepupal stage is 96 ± 2 %, with an adult emergence rate of 21 to 27 % (Tomberlin et al., 2002).

1.6. Biotransformation efficiency indicators

Indicators used to evaluate the biotransformation capacity of BSF larvae include survival rate, bioconversion rate, waste conversion efficiency, waste reduction rate, larval growth and nutrient content. Survival rate was established as a reference indicator in insect experiments by Oonincx and was subsequently used by Miranda, Myers, Lalander, Rehman and Gold (Liu et al., 2020; T. Liu et al., 2022; Miranda et al., 2020; Myers et al., 2008; Lalander et al., 2019; Rehman et al., 2017; Gold et al., 2020a). According to Gold, the survival rate increases to between 97 % and 100 % when BSF larvae are fed mixtures of different organic waste streams, compared to individual waste sources (Siddiqui et al., 2022).

The bioconversion rate measures the yield of organic waste transformation, relating the mass of the substrate to the final mass obtained by the larvae, as well as its composition in protein and fat. This rate can vary according to the composition of the feeding substrate (Gold et al., 2018; Rehman et al., 2017). Bosh and collaborators indicate that this rate should be calculated on a dry mass basis. Another important indicator is waste conversion efficiency (Z. Liu et al., 2018), also referred to as ingested/digested food conversion efficiency, which relates the amount of undigested substrate to the biomass produced (Diener et al., 2009a; Gold et al., 2020b; Oonincx et al., 2015a; Villa et al., 2021).

Given these challenges, alternative methods that promote sustainable waste valorization are needed. Among these, *Hermetia illucens* has emerged as a promising biological agent for converting organic waste into valuable products such as proteins, lipids, and biofertilizers. This study aims to evaluate the potential of BSF larvae in bioreactors for the degradation of agricultural and industrial residues under controlled conditions, providing insights into nutrient recycling and the production of sustainable bioresources.

1.7. Bioreactors and their role in process optimization

To achieve optimal rearing conditions for black soldier fly (BSF) larvae, temperatures between 27 and 33 °C, along with appropriate relative humidity and aeration processes, are necessary (Myers et al., 2008). Several authors have developed bioreactors for this purpose. For instance, Padmanabha and Streif designed a controlled environment reactor equipped with actuators for ventilation control, flow direction, lighting, and ambient temperature management, including cooling and heating. To facilitate water and nutrient exchange, they incorporated a circuit with five membrane pumps: two for gas inlet and outlet and three for water and nutrient inlet and outlet. Additionally, the system includes sensors for monitoring CO₂, O₂, and volatile organic compounds, as well as temperature and humidity. Performance tests of the sensors and microcontrollers were conducted as part of the study (Padmanabha and Streif, 2019).

Parodi and co-workers constructed 265 L open-circuit climatic breathing chambers (80 × 50 × 45 cm) with air temperature controlled at 27 ± 0.5 °C, relative humidity maintained at 70 ± 5 %, and a light (L) ratio of 1:23. The air flow rate was set at 27 L/min, and two fans were utilized to ensure adequate air mixing, resulting in air velocity varying from 0.2 to 0.8 m/s. A mass balance was calculated to quantify nutrient bioconversion efficiencies, including carbon, energy, nitrogen, phosphorus, and potassium, as well as gaseous emissions (CO₂ equivalent and ammonia) (Parodi et al., 2020).

Lalander and co-workers utilized a reactor for active ventilation control within a 3.2 m³ chamber. Ventilation is provided by four fans (Rörfläkt® ø100 mm), each with a flow rate of 107 m³/h, blowing air down through openings in the reactor chamber's roof. Additionally, two fans (Ventilationsfläkt® ø 100 mm) with a flow rate of 187 m³/h each are connected to two exhaust air pipes. The reactor chamber is equipped with combined air temperature and humidity sensors (AM2302/DHT22, Adafruit®). The study found that active aeration can contribute to substrate water depletion. Without correcting for environmental

temperature, larvae exhibited reduced growth and development (Lalander et al., 2021).

L. Cheng and co-workers developed a reactor to treat nitrogen-containing water using BSF larvae. The setup involved pumping water into the bioreactor for 1 min, followed by a 5-minute pause. The experiment was conducted in a constant temperature water bath set at 30 °C for 7 days, with each group consisting of three biological replicates. Nitrogen emissions, in the form of ammonia and N₂O, were monitored to correlate their elimination. The results indicated a reduction of nitrogen species by up to 70 % (Chen et al., 2023). Bioreactors not only optimize larval growth, but also allow monitoring of gas emissions and evaluation of nutrient conversion efficiency.

The research compared the degradation of dairy cattle manure, sheep manure, agave residues, and vinasses from mezcal production, mediated by black soldier fly larvae, in a bioreactor. The setup included monitoring of ambient temperature, relative humidity, CO₂, and pollutant gases. The growth of larvae fed the selected diets was evaluated, along with the analysis of components related to soil nutrition, as well as the production of proteins and fats in the black soldier fly larvae, with the aim of producing fertilizers and biofuels.

2. Materials and methods

2.1. Reactor design and construction at laboratory scale

The rearing process of black soldier fly larvae has specific requirements that need to be controlled, such as ambient temperature, aeration, and substrate supply. The reactor is a controlled environment model, constructed from a 50 L polystyrene container, designed to hold up to 10 vessels with a maximum treatment capacity of 3000 g of black soldier fly larvae. While the dimensions of the vessels may vary, 8.5 × 14 cm vessels were utilized.

Temperature and humidity are regulated by the STC 3028 device (operating range: −20 to 80 °C and 0 to 100 % RH), which integrates an electrical resistance heater and an ultrasonic humidifier (HUM-24 V, 15 ml/h). The air inlet and outlet are managed by a double analog timer (EBCHQ®), and three Phoenix® air fans, each with a flow rate of 170 m³/h and a speed of 3000 RPM, are included for aeration. The monitoring system consists of an ambient temperature and relative humidity sensor (DHT11), with an operating range of 0 to 50 °C and 20–95 % RH. It also includes a CO₂ sensor (CCS811) with a sensitivity range of 400 to 8192 ppm, and an air quality sensor (M135) that detects harmful gases. Additionally, four type K thermocouples (Max 6675) are employed for temperature verification in the substrate, with a measurement range of 0 to 1024 °C (Fig. 1). The programming for the system was developed using Arduino® libraries.

2.2. Variables for the elaboration of the growth model

The experimental design included feeding rates of 47.62 mg/larvae-day and 95.23 mg/larvae-day, selected according to the study by Meneguez et al. (Meneguez et al., 2018). The diets used to evaluate the growth of black soldier fly (BSF) larvae consisted of bovine feces, ovine feces, and mezcal process residues (vinasse and bagasse in proportions of 40:60 and 50:50). Growth in mass and length was measured over a 21-day period. The environmental temperature in the reactor was maintained between 28 and 30 °C, with relative humidity set at 55–60 %. Three replicates were performed for each experimental condition to ensure the reliability and reproducibility of the results.

Based on assays conducted in previous research, survival, protein conversion, bioconversion, and percentage reduction rates were evaluated using Eqs. (1)–(5) from the following studies: (Diener et al., 2009b; Gold et al., 2020b; Liu et al., 2022; Parodi et al., 2021; Villa et al., 2021; Wang and Wang, 2016; Zhao et al., 2009)

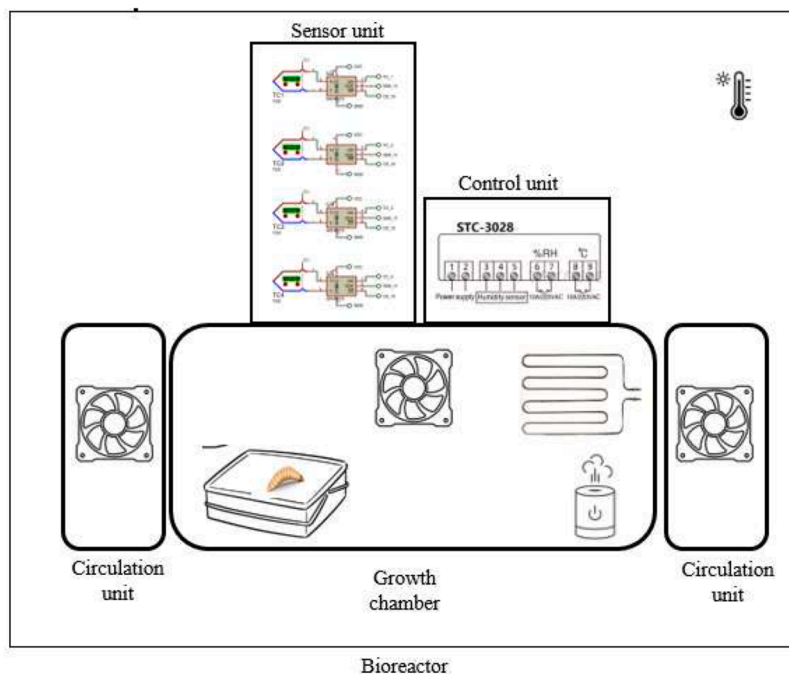


Fig. 1. Bioreactor at laboratory scale.

$$\text{Survival rate (\%)} = \frac{\text{Larvae}_{\text{end}}}{\text{Larvae}_{\text{beg}}} * 100\% \quad (1)$$

Larvaend: number of larvae at the end of experiment. Larvaebeg: number of larvae at the beginning of the experiment.

$$\text{Bioconversion rate}(\%DM) = \frac{\text{larvae}_{\text{gain}}(g)}{\text{feed}_{\text{mass}}(g)} * 100 \quad (2)$$

Larvaegain: mass of the larvae at the end of experiment (g). Feedmass: mass of feed supplied to larvae (g)

$$\text{Protein conversion efficiency (\%DW)} = \frac{\text{Protein}_{\text{acc}}(g)}{\text{feed}_{\text{mass}}(g)} * 100 \quad (3)$$

Proteinacc: Protein accumulation in larvae (g) . Feedmass: mass of feed supplied to larvae (g)

$$\text{Waste conversion efficiency (\%DW)} = \frac{\text{larvae}_{\text{gain}}(g)}{\text{feed}_{\text{mass}}(g) - \text{residue}_{\text{mass}}(g)} * 100 \quad (4)$$

Larvaegain: mass of the larvae at the end of experiment (g). Feedmass: mass of feed supplied to larvae (g). Residuemass: mass of residue at the end of experiment (g).

$$\text{Waste reduction (\%DW)} = \left[1 - \frac{\text{residue}_{\text{mass}}(g)}{\text{Media DW}(g)} \right] * 100 \quad (5)$$

Residuemass: mass of residue at the end of experiment (g). Media DW: substrate in dry mass at the beginning of the experiment.

2.3. Chemical composition of substrates and larvae

2.3.1. Lignin determination method

Lignin is used in the Monod growth equation as a reference for the transformation of substrate to biomass and as a basis for establishing Ks (Wang and Witarsa, 2016). In the method used for lignin determination, samples of dry substrate and biomass (frass) were dried to a constant mass. Samples of 1 g were taken, and hydrolysis was carried out with 0.1 M sulfuric acid in a water bath at 40 °C for 2 h. The mixture was then filtered under vacuum through a previously dried filter, which was

weighed at 105 °C for 2 h. The percentage of lignin was obtained by the gravimetric method.

2.3.2. Total Nitrogen - protein content determination method (Kjendahl)

The equipment used for digestion was the Turbotherm, and for distillation, the Vapodest 30S from Gerhardt Analytic®, utilizing the Kjeldahl method. For the substrate and frass samples, calcination was performed at 550 °C. The larvae samples were not calcined, and 0.5 g samples were weighed. In a digestion tube, 2 g of selenium catalyst reagent and 20 ml of concentrated sulfuric acid were added. The digestion was carried out at 420 °C until the sample became transparent.

After digestion, the samples were distilled by adding 20 ml of 0.1 M sodium hydroxide to the digestion tube. In the Erlenmeyer flask, 10 ml of boric acid solution, along with methyl red and bromocresol green indicators, were added and titrated with 0.01 N sulfuric acid until the solution changed from green to pink (Jiang et al., 2014). Eq. (6) was used to determine the total nitrogen concentration.

$$\text{Total nitrogen : } \frac{(T - B) * N * 14}{S} \quad (6)$$

T: volume of titrant spent on the sample. B: volume of titrant spent on blank. S: weight of sample. N: normality of acid.

Let T be the ml of titrated sulfuric acid used for the sample, B be the ml of titrated sulfuric acid used for the blank, N be the exact standardization of the sulfuric acid, and S be the sample weight. The standardization of the sulfuric acid used for volumetric analysis was performed with a Na₂CO₃ solution. The adjustment is carried out using Eq. (7):

$$C = \left(\frac{m\text{Na}_2\text{CO}_3 * P\text{Na}_2\text{CO}_3 * 1000}{M\text{Na}_2\text{CO}_3 * (V1 - V0)} \right) \quad (7)$$

Let C be the molar concentration of the H₂SO₄ solution (mol/L), m be the mass of Na₂CO₃ (g), P be the purity of the Na₂CO₃ reagent (mass fraction), V1 be the volume of H₂SO₄ used in the titration of the Na₂CO₃ standard (ml), and V0 be the volume of H₂SO₄ used in the titration of the blank (ml) (Giraldo Gómez et al., 2017).

2.3.3. Method for lipid determination

The determination of fat content was carried out using Soxtherm®

equipment, following the gravimetric method: the test tubes were dried for 2 h at 105 °C, cooled in a desiccator, and weighed. Then, 150 ml of ethyl ether was added to the cartridge inserted into the test tube. The equipment was programmed for 4 extraction cycles. Afterward, the test tube was placed in an oven at 105 °C for one hour, cooled, and the mass difference was determined (Hewavitharana et al., 2020).

2.3.4. Colorimetric method for total organic carbon determination

The calibration curve was prepared using glucose concentrations of 1.5, 4, 10, 20, and 40 in 100 ml beakers. Then, 10 ml of 2 N potassium dichromate solution was added and allowed to react for 5 min, followed by the addition of 10 ml of concentrated sulfuric acid. The mixture was left to cool in the extraction hood for three hours. The same procedure was applied to the samples. Subsequently, 50 ml of deionized water was added, and the samples were analyzed using a spectrophotometer at 635 nm (ICONTEC, 2022).

The concentration of oxidizable carbon in solid products was determined using the following Eq. (8).

$$\% \text{oxidizable carbon} = \left(\text{lecture} \times RF \times \left(\frac{0.1}{Wm} \right) \right) \quad (8)$$

RF: reference material factor. 0.1: reference value for the weighed mass of glucose, equivalent to 40 % oxidizable organic carbon. Wm: sample weight in g.

2.3.5. Colorimetric method for phosphorus determination. Olsen method (Recena et al., 2022)

A calibration curve was created with phosphorus concentrations of 0.1, 0.2, 0.4, and 0.8 ppm, prepared from a standard solution of monobasic potassium phosphate. For the samples, 0.9999 g were weighed, 7 ml of extractive solution were added, and the mixture was shaken for 30 min at 180 oscillations per minute. The samples were then filtered using Whatman No. 42 filter paper or another similar grade. Aliquots of 1 ml were taken, to which 0.5 ml of molybdenum reagent and 0.3 ml of ascorbic acid solution were added, and then the volume was brought up to 6 ml with deionized water. The absorbance was measured at 882 nm. The concentration of total phosphorus was determined using Eq. (9).

$$P(\text{mg} / \text{Kg}) = \left(CC \times \left(\frac{Vi}{P} \right) \times \left(\frac{Vf}{a} \right) \right) \quad (9)$$

CC = mg/L of phosphorus in the solution, Vi = volume of extractant solution added, P = weight of the air-dried sample, Vf = final volume of the colorimetric solution to be measured, a = aliquot of the sample used for quantification.

2.3.6. Method for determination of Ca, Mg. Atomic absorption

AA200Analyst® Atomic Absorption equipment was used for calcium and magnesium analysis by atomic absorption technique with hollow cathode lamp.

3. Results and discussion

Hermetia illucens larvae were supplied by the Universidad Autónoma de Querétaro. The larvae were 5 days old and had been fed chick feed since hatching.

The selected substrates for feeding the BSF larvae included bovine feces, ovine feces (obtained from CITAP located in Atlixco, Universidad Popular Autónoma del Estado de Puebla), and two mixtures of agave bagasse from the *potatorum* species and vinasse from the mezcal production process (Escuela de Mezcal de Puebla).

3.1. Moisture of the substrate during the experiment

The required moisture conditions for the substrate must exceed 50 %

(Cheng et al., 2017). For this reason, the experiments conducted with the agave and vinasse samples required the addition of water on day 11, while the sheep feces substrate experiment, with a feeding rate of 95.23 mg/larvae.day, necessitated an increase in moisture on day 15. During the experiments, the moisture levels of the cattle and sheep feces substrates with feeding rates of 47.61 mg/larvae.day remained constant. The bovine and ovine feces substrates with feeding rates of 95.23 mg/larvae.day exhibited the highest moisture levels, with only minor fluctuations (Fig. 2).

Hbo.T1: bovine manure, feeding rate 95.23 mg/larvae.d. Hbo.T2: bovine manure, feeding rate 47.61 mg/larvae.day. HO.T1: sheep manure, feeding rate 95.23 mg/larvae.day. HO.T2: sheep manure, feeding rate 47.61 mg/larvae.day. AV60.40.T1: agave bagasse and mezcal vinasse in a 60:40 ratio, feed dose 95.23 mg/larvae.day. AV60.40. T2: agave bagasse and mezcal vinasse in ratio 60:40, feed rate 47.61 mg/larvae.day. AV50.50.T1: agave bagasse and mezcal vinasse in 50:50 ratio, feed rate 95.23 mg/larvae.day AV50.50.T2: agave bagasse and mezcal vinasse in 50:50 ratio, feed rate 47.61 mg/larvae.day.

The ideal food moisture content for BSF larvae is in the range of 70 to 80 %, while the lower threshold of that value is likely to be between 40 and 55 % (Dortmans et al., 2017; Ermolaev et al., 2019; Lalander et al., 2019; Zhang, 2013).

3.2. Larval growth (mass and length)

The highest mass was obtained in larvae fed with sheep feces, TA 95.23 mg/larvae.day, until day 19. Larvae fed with mixtures of agave bagasse and vinasse maintained their low weight throughout the experiment (Fig. 3).

Larvae with the highest growth in length were fed TA 95.23 mg/larvae.day, achieving an average growth of 1.54 cm above 1.16 reported by (Julita et al., 2018) Whereas, larvae fed with bagasse:vinasse mixtures did not exceed 1 cm in length, it is likely that these mixtures contain low nutrient concentrations, slowing growth (Fig. 4).

3.3. Characterization of substrates and organic fertilizers

In the European Union, an organic solid fertilizer is defined as material containing at least 1 % nitrogen or a sum of NPK content of at least 4 %. The NPK ratio in organic fertilizers, as established by Regulation (EU) 2019/, is 10:4:6 (Fernández, 2022). According to standards in Asia, the nitrogen content in fertilizers should be higher than 1.5 % in Malaysia, 4 % in South Korea, and between 2.5 % to 5 % in the Philippines (Abd Manan et al., 2024). In the United States, plant or animal fertilizers must comply with a C:N ratio between 25:1 and 40:1 (Makan, 2015). In Colombia, the recommended nitrogen content is at least 1 %, with a C:N ratio of less than 25:1 (Bohórquez et al., 2015). However, the NPK ratio for black soldier fly larvae feces, as mentioned by Gärttling & Schulz, corresponds to 1:0.9:1.1 (Gärttling and Schulz, 2022).

The results of the characterization of the substrates, larvae BSF and fertilizers obtained are presented in Tables 1–3.

In the characterization performed by Gold on bovine manure, the moisture content was 87 %, and the protein content was 11.1 % (Gold et al., 2020a), with a conversion factor to protein of 4.3 (Cheng et al., 2017). In the present study, the mean protein concentration for bovine feces was found to be 1.075 %. This reduction may be due to outdoor storage and climatic factors that decrease the nitrogen content in feces. In the experiment with bovine and ovine feces, nitrogen showed a slight increase after the biotransformation process. Furthermore, Oonincx et al. found that high-protein diets contained twice as much phosphorus as low-protein diets (Oonincx et al., 2015b), suggesting that improving the protein content in the substrate could improve feces conditions.

In our research, the C:N ratio in the compost obtained from the degradation of bovine feces was 18.75, 340 while for sheep feces at a feeding rate of 95.23 mg/larvae.day, it was 10.10. In the degradation of

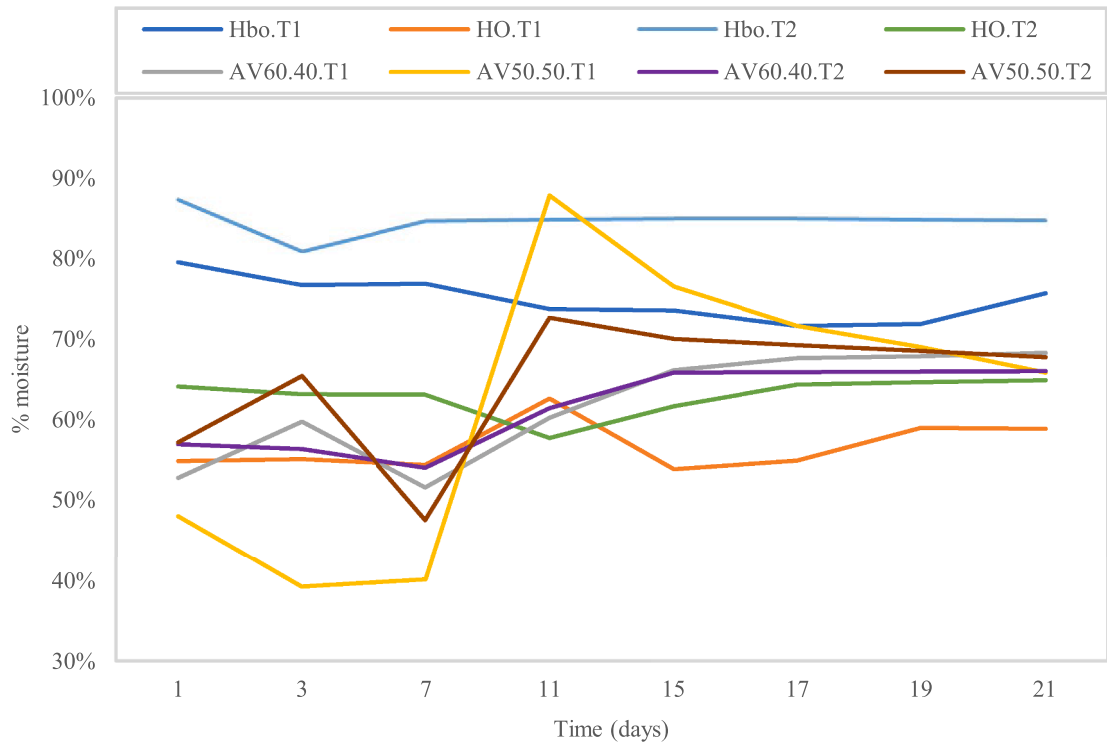


Fig. 2. Substrate moisture behavior during biodegradation.

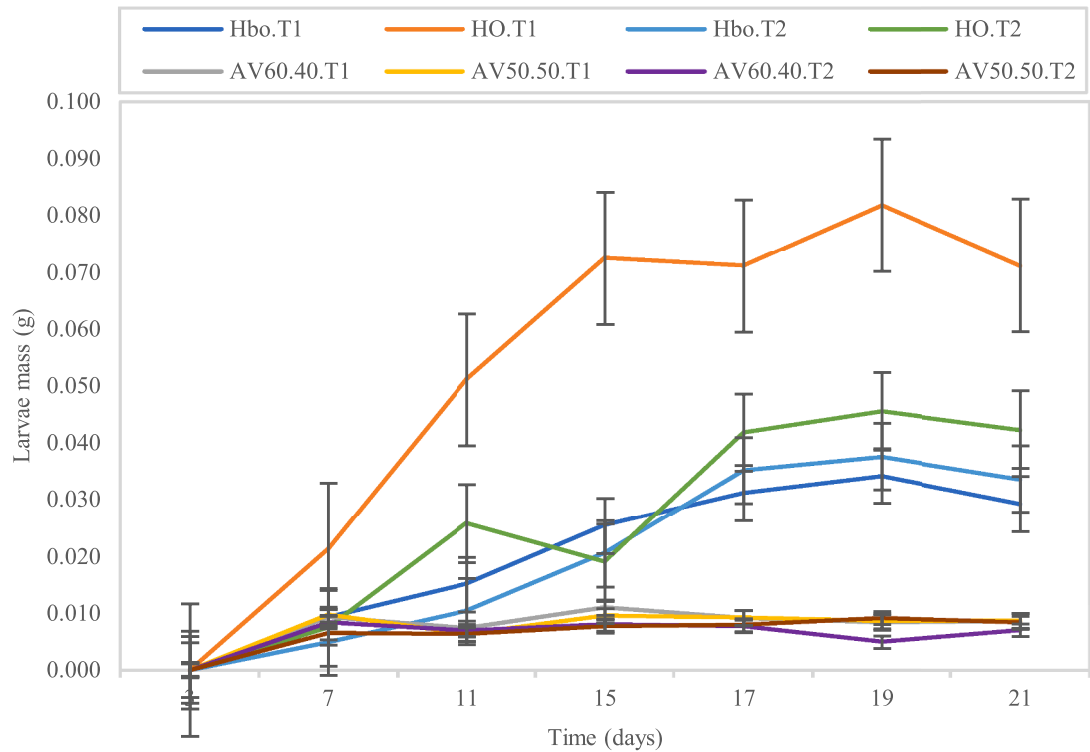


Fig. 3. Mass growth (g) of larvae.

agave mixtures and 341 mixtures of agave and bagasse, the C:N ratio exceeded 60:1, indicating that complete 342 degradation of the substrate was not achieved. Compost produced from sheep feces at a feed rate of 95.23 mg/larva.day met the 343 requirement of having more

than 1 % nitrogen, although it contained very low concentrations of phosphorus.

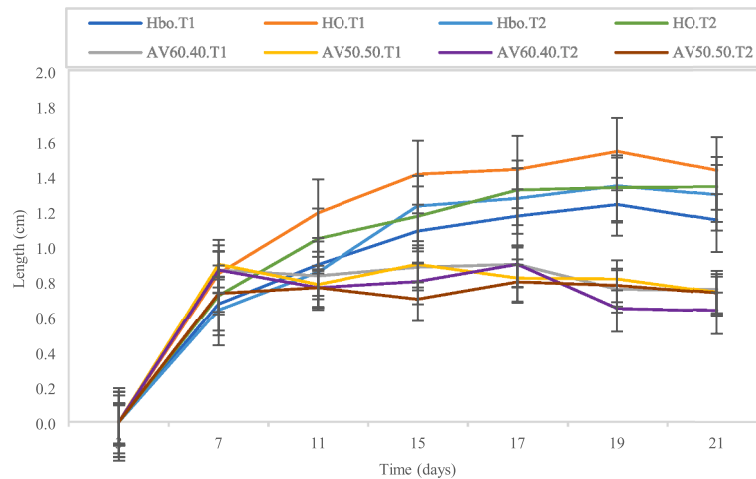


Fig. 4. Growth in length (cm) of larvae.

Table 1

Characterization of substrates (Bovine and Ovine Feces) and frass obtained in the bioconversion.

Sustrate- biomass	bovine manure		sheep manure		Frass larvae- bovine HboT1	Frass larvae- bovine HboT2	Frass larvae-sheep HOT1	Frass larvae-sheep HOT2
Lignin (%)	2.73	±0.03	5.445	±0.11	0.95	14.75	2.04	23.06
Moisture (%)	83.62	±5.25	59.29	±6.83	74.93	84.75	59.74	64.89
Ashes (%)	5.83	±2.21	9.2	±2.59	9.52	3.12	18.13	9.78
Nitrogen (%)	0.25	±0.23	0.46	±0.33	0.32	0	1.3	0.32
Total organic carbon (%)	2.55	0	16.91	±4.17	6.06	3.92	13.14	10.64
Phosphorus (mg/ Kg)	0.022	±0.01	0.028	±0.003	0.006	0.023	0.024	0.030
Calcium (%)	0.62	±0.01	0.57	±0.01	0.58	0.64	0.65	0.59
Magnesium (%)	0.635	±0.01	0.645	±0.01	0.66	0.63	0.63	0.61

Table 2

Characterization of substrates (agave bagasse and vinasas) and frass obtained in bioconversion.

Sustrate- biomass	bagasse agave	vinasse	bagasse agave vinasse 40:60	bagasse agave vinasse 50:50	Frass bagasse agave vinasse AV60.40 .T1	Frass bagasse agave vinasse AV60.40 .T2	Frass bagasse agave vinasse AV50.50 .T1	Frass bagasse agave vinasse AV50.50 .T2
Lignin (%)	13.88	–	–	–	14.36	15.49	14.59	14.94
Moisture (%)	13.91	93.84	52.58	54.88	68.31	66.06	65.86	67.74
Ashes (%)	21.06	1.03	11.11	12.53	10.47	13.65	11.68	12.16
Nitrogen (%)	0	0.18	–	–	0.22	0.11	0.08	0
Total organic carbon (%)	9.25	–	–	–	14.19	6.85	17.26	13.22
Phosphorus (%)	0.0004	–	0.001	0.001	0.011	0.019	0.010	0.010
Calcium (%)	2.42	–	1.17	1.12	–	–	–	–
Magnesium (%)	2.51	–	1.25	1.22	–	–	–	–

Table 3

Composition of BSF larvae.

Experiment	Larvae bovine Hbo.T1	Larvae sheep HO.T1	larvae bovine Hbo. T2	Larvae sheep HO.T2	Larvae bagasse- agave vinasse AV60.40. T1	Larvae bagasse- agave vinasse AV50.50. T1
Moisture (%)	72.15	66.27	81.97	76.62	74.94	71.05
Nitrogen (%)	0.83	2.33	1.03	3.77	0.42	1.14
Protein (%)	5.20	14.60	6.43	23.54	2.76	7.42
Lípidos (%)	6.82	9.73	–	–	0.72	0.52

3.4. Biotransformation indicators

Miranda et al. published those larvae fed dairy cow manure in an industrial-scale environment experienced stunted growth and had lower survival rates compared to those fed pig and poultry manure. The presence of cellulose, hemicellulose, and lignin in cattle manure makes it more difficult to degrade, which could hinder the conversion efficiency of the larvae (Miranda et al., 2020). However, Myers indicated that the survival rate of BSF larvae in bovine manure is greater than 70 % at the laboratory scale (Myers et al., 2008). In Gold et al.'s research, the bioconversion rate for bovine manure was found to be 3.8 % (Gold et al., 2018). Research by Rehman and collaborators demonstrated that mixing manure from different livestock species can aid in the degradation of complex fibers. In a 1:1.5 mixture of dairy cattle and poultry

manure, better larval biomass yields, feed conversion rates, and nutrient reduction were achieved (Rehman et al., 2017). For our experiment, the highest survival rates, protein conversion, bioconversion, waste conversion efficiency, and waste reduction were observed in larvae fed on sheep feces. The low bioconversion rate of *agave* bagasse is likely due to the presence of calcium oxalate, which acts as an irritant (Rosas Hernández, 2015; Barabash-Neila et al., 2011), causing rejection by the larvae. Table 4 includes information on biotransformation rates of substrates and biomasses.

An analysis of variance was performed on the bioconversion rate for the experiments, the result shows that bioconversion differs due to feeding rates and diet type, rejecting the null hypothesis, included in Table 5.

3.5. Validation of sensors in the bioreactor

3.5.1. Temperature and humidity sensors

To validate the behavior of the sensors installed in the bioreactor, the *agave*-vinasse bagasse test was conducted in Puebla, Mexico, where the minimum ambient temperature ranged from 9 °C to 22 °C. The sensors were programmed to collect data every minute, including a humidity and temperature sensor (DHT11), a CO₂ sensor, an air quality sensor (M135), and four thermocouples inserted in a replica of the second trial.

According to the manufacturers of the sensors, heating of the system is necessary. Therefore, the STC 3028 device was programmed to turn the heater on below 27 °C and off at 29 °C. Additionally, the humidifier was programmed to activate below 50 % and deactivate at 65 % RH. The operation of the DHT11 sensor was verified, and the growth chamber reached a temperature of 27 °C by minute 133.

During the 21-day trial, the DHT11 sensor recorded an average temperature of 28.1 °C, with a minimum of 20.2 °C and a maximum of 34.7 °C, effectively maintaining the necessary conditions for larval growth (Fig. 5).

The K-type thermocouples (Max 6675) monitored the temperature of the substrates, which reached and maintained the target temperature after 16.5 h. It was noted that the temperature in the growth chamber decreased due to the reduction in ambient temperature at night.

The highest substrate temperature recorded was 34.5 °C in the AV60:40 T2 substrate, followed closely by AV50:50 T1 at 34 °C, AV60:40 T1 at 33.5 °C, and AV50:50 T2 at 31.5 °C, all observed between days 10 and 11 of the experiment. In research by Bonelli et al., a constant reactor temperature of 27° ± 0.5 °C was maintained, but the substrate temperatures increased to 40 °C (Bonelli et al., 2019). Kim et al. reported that temperatures can rise up to 45 °C in black soldier fly larvae due to enzyme production for protein degradation (Kim et al., 2011). Similarly, Parra Paz et al. suggested that larvae tend to cluster within the substrate, which raises the temperature due to overcrowding and movement (Parra Paz et al., 2015).

The moisture in the growth chamber remained consistently between 50 % and 65 % during the first two weeks, but decreased to 35 % in the third week. However, this decrease in moisture in the growth chamber did not appear to affect the moisture content in the substrates, which ended with an average of 66 %.

3.5.2. Gas sensors

The bioreactor was programmed to aerate the growth chamber every 60 min by running the fans for five minutes. During the first 24 h, the average CO₂ concentration was 3394 ppm, while the air quality index was 285 ppm. The highest concentration recorded was 18 433 ppm CO₂ on day 4, and 466 ppm for air quality on day 21 of the experiment (Fig. 6). After day 11, the CO₂ concentration remained in the upper detection range. Although the recommended CO₂ concentration for humans in closed enclosures is less than 800 ppm (C. Wang et al., 2024), this study cannot determine whether the CO₂ concentration had any impact on the low growth of larvae fed with *agave* bagasse and vinasse. However, it may serve as an adjustment factor for aeration in future experiments.

4. Conclusions

The bioreactor, designed for controlling and monitoring environmental variables and self-developed gases, was programmed with open-source software. It was developed and tested through an experiment that involved feeding black soldier fly larvae with waste cattle manure, sheep manure, and mixtures of *agave* bagasse and vinasse. The trials were conducted with a growth chamber temperature set between 27 °C and 29 °C, and relative humidity maintained between 50 % and 65 %. The larvae fed with manure exhibited growth similar to that reported by other authors using the same diet.

The growth chamber was programmed to aerate every 60 min for 5 min; however, the CO₂ concentration exceeded the levels recommended for human occupancy in closed environments. This result can be utilized to establish aeration conditions based on the CO₂ concentration in the air. Additionally, the system enabled the monitoring of substrate temperature changes and demonstrated a low incidence of relative humidity affecting the moisture content in the substrates.

The system allowed access to recorded data and adjustments of parameters in the controller. The reactor was designed with a scalability approach for areas with low temperatures or significant seasonal temperature changes, ensuring the maintenance of a continuous process. The results generated in this study can be applied to biomass production, the creation of by-products, and the investigation of growth responses to various substrates and environmental factors in controlled environments. Furthermore, this study facilitated the collection of fundamental data for the development of mathematical models, which will be the focus of future work.

The use of *Hermetia illucens* and bioreactors for managing and valorizing organic and agro-industrial waste represents an innovative and promising approach that combines effective waste reduction with the production of high-value biomass. However, the large-scale implementation of these technologies faces significant challenges. One key concern is energy consumption associated with the control of critical parameters such as temperature, humidity, and ventilation in bioreactors, which raises questions about the sustainability of the process. While the use of sensors and automated systems enhances precision and control, it also potentially increases indirect carbon emissions due to the energy required to operate these devices.

Another major limitation is the dependence on controlled laboratory

Table 4
Biotransformation rates of substrates and biomasses.

Experiment	Larvae bovine Hbo. T1	Larvae sheep HO. T1	larvae bovine Hbo. T2	Larvae sheep HO. T2	Larvae AV60.40.T1	Larvae AV50.50.T1	Larvae AV60.40.T2	Larvae AV50.50.T2
Survivor rate (%)	89.78	89.92	74.22	92.42	88	87	57	64
Bioconversion rate (%)	2.63	6.98	4.75	6.36	0.70	0.63	0.14	0.26
Protein conversion (%)	8.50	10.78	16.92	21.86	3.3	5.70	–	–
Waste conversion (%)	16.06	8.62	–	32.90	2.93	3.11	1.41	0.71
Waste reduction (%)	35.63	57.38	11.64	34.73	43.04	37.09	23.19	52.92
Waste reduction (%/day)	1.70	2.73	0.55	1.65	2.05	1.77	1.10	2.52

Table 5
Statistical analysis of bioconversion.

Source of Variation	SS	Df	MS	F	P-value	F crit
Between Groups	0.0109659	5	0.00219318	6.07433912	0.00410158	3.0254383
Within Groups	0.00469374	13	0.00036106			
Total	0.01565964	18				

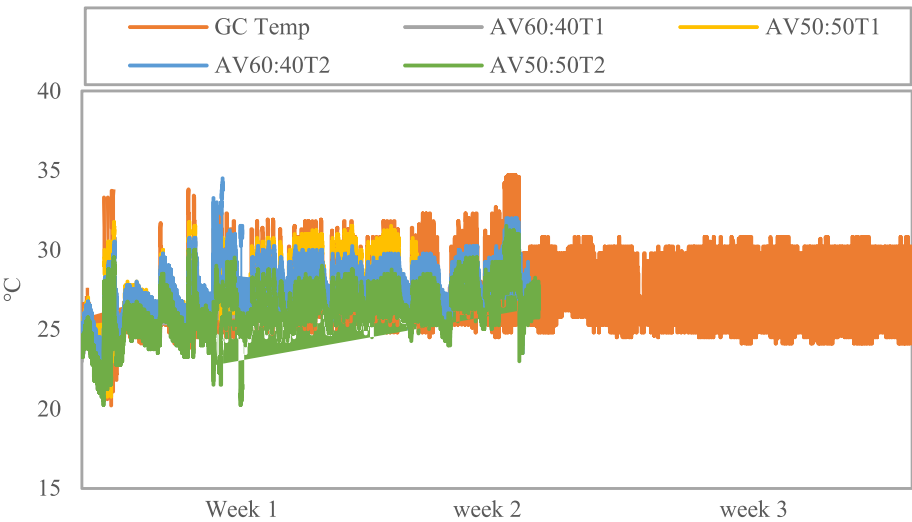


Fig. 5. Behavior of the growth chamber temperature and substrate temperature.

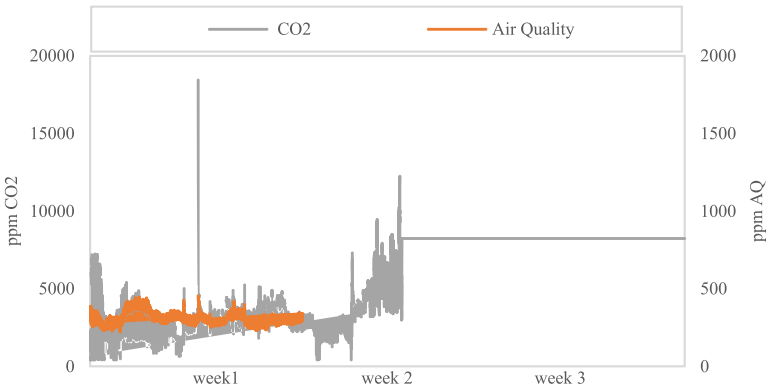


Fig. 6. Concentration of gases in the growth chamber.

conditions, such as temperature, humidity, and ventilation, which may differ substantially from those encountered in large-scale applications. These differences could affect the reproducibility of results in real-world or industrial environments. Moreover, the specific types of waste evaluated, the proportions used in the study may not reflect the wide range of organic and agro-industrial residues available, limiting the broader applicability of the findings to other types of waste with different physicochemical characteristics.

Additionally, it is crucial to strike a balance between operational energy costs and the environmental benefits of technology. Careful evaluation of this balance will ensure both the economic and environmental viability of the process. The adoption of renewable energy sources and the optimization of bioreactor designs could help mitigate

these issues, reducing the carbon footprint of the process. These strategies, along with ongoing research to improve system efficiency, are vital for ensuring the long-term sustainability of this approach and its potential contribution to the circular economy and waste management

CRedit authorship contribution statement

Diana Marcela Cuesta-Parra: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Carlos Enrique Montenegro-Marín:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Genoveva Ortega-Rosano:** Writing – review & editing, Validation,

Supervision, Resources, Project administration, Funding acquisition, Formal analysis. **Francisco Javier Sanchez-Ruiz:** Writing – review & editing, Validation, Supervision, Software, Methodology, Formal analysis, Data curation.

Declaration of competing interest

The authors certify that they have no conflict of interest for the publication of the submitted article.

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Data availability

Data will be made available on request.

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